VOLUME 07 / 开源工程

开源Agent仓库与工程图谱

36个仓库的架构定位、适用场景、风险、生态关系与选型路径

研究范围	2025-08-20 至 2026-08-19; 另列少量边界基线
语料规模	GitHub元数据快照: 2026-08-19 (America/Los_Angeles; GitHub metadata may show 2026-08-20 UTC); 共 36 个仓库
证据原则	优先论文原文、作者项目页、官方工程博客和GitHub仓库; 主张与推断分开
版本	v1.0 | 生成日期 2026-08-19

本报告不是排行榜。它是一套研究地图:
解释哪些结果可以相信、哪些只是方向性证据、哪些工程选择会改变结论。

本卷导航

1. 仓库生态的七类分层

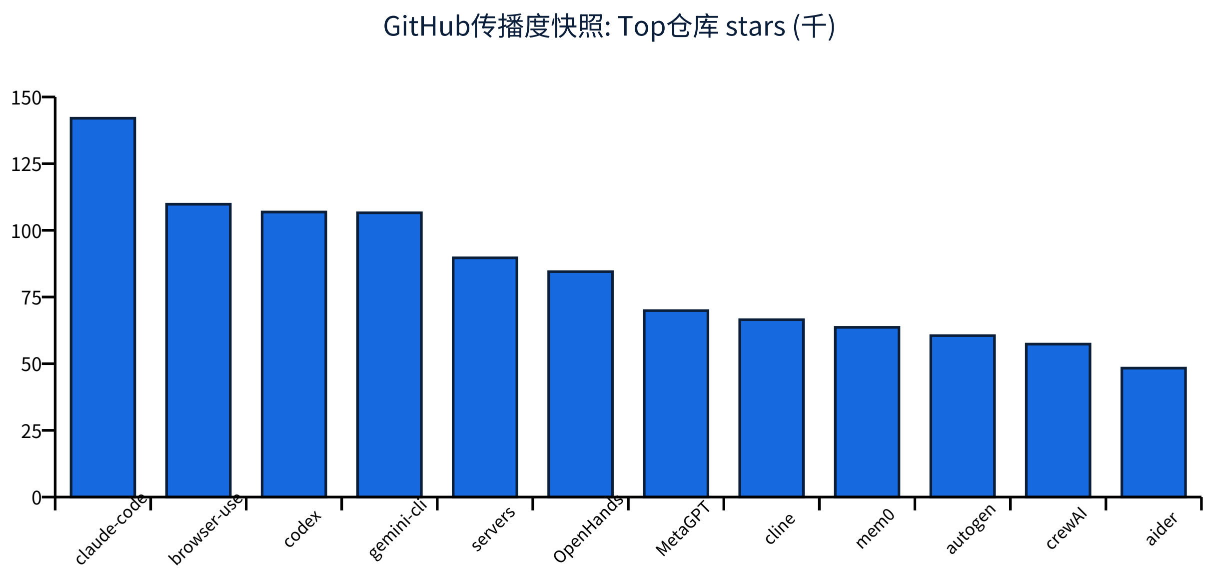
开源Agent仓库不能放在一个排行榜里。SDK、耐久运行时、完整harness、产品应用、协议、记忆层和训练基础设施解决不同问题。选错层会导致团队用一个工具填补它没有承诺覆盖的生产能力。

层	代表	负责	[illegible]
轻量SDK	openai-agents, PydanticAI, smolagents	模型/工具循环、类型和最小抽象	原型和嵌入应用
耐久运行时	LangGraph, ADK	状态、事件、恢复、人机协作、部署	生产工作流
完整harness	Claude Agent SDK, DeepAgents	电脑、文件、子Agent、压缩、计划	长时通用任务
多Agent框架	AutoGen, CrewAI, CAMEL, MetaGPT	通信、角色、任务和拓扑	协作研究/流程
领域应用	Codex, Claude Code, Gemini CLI, OpenHands, browser-use	编码或浏览器端到端体验	直接工作产品
状态/记忆层	mem0, Graphiti, Letta	跨会话事实、图、长寿命Agent状态	需要持久性产品
协议/训练/环境	MCP, A2A, Agent Lightning, Orchard	互操作、RL、rollout和评测环境	平台和研究基础设

两个常见误区

- “框架提供memory”不等于完成了业务记忆设计。仍需定义schema、来源、权限、过期、删除和评测。
- “支持MCP/多Agent/沙箱”不等于默认配置安全可靠。必须检查实现边界、部署模式和工具来源。

2. 传播度、成熟度与误读



stars只反映传播与关注，不代表可靠性、安全性或适配性。

Stars的使用方法

Stars可用于衡量开发者关注、教程生态和招聘熟悉度，但受品牌、发布日期、产品热度、营销和迁移影响。它不能直接衡量提交质量、故障率、API稳定、安全或适配性。本报告只把stars作为快照字段，并把架构和风险单独分析。

仓库成熟度应看什么

维度	检查	原因
维护	最近提交、发布节奏、bus factor、issue处理	避免只看总commit
兼容	Python/Node版本、模型提供商、协议版本	Agent生态变化快
质量	单测、集成/e2e、类型、CI、可复现样例	README demo不够
运行	持久化、并发、恢复、超时、取消、幂等	长时系统核心
安全	沙箱、网络、秘密、权限、供应链	默认值比功能列表更重要
可观测	trace、事件、成本、状态diff、回放	没有轨迹难以运营
治理	许可证、贡献、漏洞披露、路线图	企业采用前置条件

快照限制

GitHub计数是2026-08-19 PT附近的动态快照，可能在用户下载时已经变化；API时间以UTC显示。归档状态也可能意味着迁移到新仓库，而非项目死亡，必须检查官方迁移说明。

3. 核心框架比较矩阵

项目	类型	语言	许可	设计中心	适合	关
openai-agents-python	SDK/runtime	Python	MIT	Small agent loop with agents, tools, handoffs, sessions, guardrails and tracing.	OpenAI-centric service agents and multi-agent workflows.	Provider coupl moving platfor version pinning
claude-agent-sdk-python	SDK/harness	Python	MIT	Generalizes Claude Code's harness primitives to application agents.	Agents that need a computer, subagents, hooks and permission controls.	Closely tied to and platform e
adk-python	SDK/runtime	Python	Apache-2.0	Code-first agents, tools, sessions, memory, artifacts, evaluation and deployment.	Google Cloud/Gemini teams needing multi-agent and managed deployment.	Broad abstract obscure runtim ownership.
autogen	multi-agent framework	Python	CC-BY-4.0 metadata	AgentChat abstractions over an event-driven core and extensions.	Research and conversational multi-agent orchestration.	Large version t licensing/comp review.
langgraph	durable runtime	Python	MIT	Graph/state runtime with durable execution, persistence, streaming and human-in-the-loop.	Stateful production workflows requiring recovery and explicit control flow.	Graph complex application-sp debt.
deepagents	agent harness	Python	MIT	Batteries-included harness adding planning, file state, subagents and memory on LangGraph.	Long-horizon research and coding prototypes.	Convenience la dependencies the LangChain
crewAI	multi-agent framework	Python	MIT	Role/task/crew and flow abstractions for collaborative automation.	Business workflows where roles and sequence are easy to express.	Role-playing ca unless informa responsibility b
pydantic-ai	typed SDK	Python	MIT	Typed model and tool interfaces integrated with validation, evals and observability.	Python teams prioritizing type safety and model portability.	A typed bound errors, not sem failures.
smolagents	minimal SDK	Python	Apache-2.0	Minimal agents that often express plans and tool composition as code.	Readable experiments and code-agent research.	Small core doe production con isolation by its
browser-use	web agent	Python	MIT	Browser abstraction plus agent loop over Playwright-like interactions.	Web automation and browser-agent prototyping.	Site change, ar credentials and create operatio
OpenHands	coding platform	TypeScript	MIT	Full coding-agent platform with isolated runtime, sessions, UI and extensible agents.	Multi-user hosted software-development agents.	Deployment co larger than a lo
letta	stateful agent platform	mixed	Apache-2.0	Long-lived agents with editable memory, tools and state management.	Research and products centered on persistent agent identity/state.	Long-term stat drift and lifecy

框架选择不是功能打勾

功能表会让成熟项目看起来都支持工具、memory、多Agent和trace。真正差异在控制模型: LangGraph强调显式状态图和耐久执行；OpenAI Agents SDK强调轻量Agent/handoff/session/trace；ADK覆盖构建到部署；Claude Agent SDK提供电脑和长时harness原语；PydanticAI强调类型；smolagents强调小核心和代码动作；AutoGen/CrewAI/CAMEL侧重协作抽象。团队应选择与自身控制需求同构的框架，而不是功能最多者。

供应商耦合的正确看法

与模型供应商深度耦合可以获得最佳原生工具、推理提示、缓存和托管能力，代价是行为与API变化。模型无关框架提高可替换性，但最低公分母可能损失模型特定能力，并把兼容测试转移给用户。可行架构是业务状态和工具协议中立，模型适配层显式版本化，harness保留模型特定优化。

4. Coding与Computer Use仓库

仓库	stars	语言	架构重点	适合	
openai/codex	106,864	Rust	Terminal coding harness with sandboxed command execution, repository context and iterative edits.	Local and remote software engineering workflows.	Model-specific b operational surf harness, not mo
anthropics/claude-code	142,023	Python	Terminal agent with file, shell, git, subagent, hook, skill and compaction primitives.	Long-running coding and general computer work.	Client repositor expose the full h control plane.
microsoft/magentic-ui	10,069	Python	Human-centered browser and file agent built on AutoGen/Magentic patterns.	Research on oversight and interactive web automation.	Live web fragilit security remain
browser-use/browser-use	109,782	Python	Browser abstraction plus agent loop over Playwright-like interactions.	Web automation and browser-agent prototyping.	Site change, ant credentials and create operatio
OpenHands/OpenHands	84,512	TypeScript	Full coding-agent platform with isolated runtime, sessions, UI and extensible agents.	Multi-user hosted software-development agents.	Deployment co larger than a loc
SWE-agent/SWE-agent	20,079	Python	Research harness for GitHub issue resolution and other terminal tasks.	Transparent coding-agent research and benchmark baselines.	Benchmark perf sensitive to env contamination.
Aider-AI/aider	48,333	Python	Git-aware terminal pair programmer with repository maps and edit formats.	Interactive coding with explicit human steering.	Less suited to u multi-user contr workloads.
cline/cline	66,499	TypeScript	IDE, CLI and SDK agent with file, terminal, browser and approval flows.	Developer-in-the-loop work inside editors.	Extension perm configuration ex surface.
continuedev/continue	35,546	TypeScript	Open coding agent and editor integrations with model/provider flexibility.	Teams needing self-hosted or model-flexible coding assistance.	Configuration fl integration and the operator.
google-gemini/gemini-cli	106,586	TypeScript	Terminal agent with Gemini models, tools, MCP and extensions.	Google-centric development and general terminal workflows.	Provider coupli terminal permis containment.
BloopAI/vibe-kanban	27,856	Rust	Task-board orchestration over multiple external coding agents.	Human supervision of parallel coding work.	It coordinates h not eliminate m context problem
humanlayer/humanlayer	11,296	TypeScript	Human-in-the-loop workflow and context engineering for complex codebases.	High-context changes needing planning and review checkpoints.	Process quality disciplined task review capacity

本地CLI、IDE扩展和托管平台

形态	代表	优势	风险
本地CLI	Codex/Claude Code/Gemini CLI/Aider	低延迟接近仓库、用户直接监督	主机权限、秘密和环境差异
IDE扩展	Cline/Continue	交互上下文和审批体验好	扩展权限与编辑器状态
完整平台	OpenHands	多用户、会话、沙箱和UI	部署、隔离、队列和存储复杂
浏览器框架	browser-use/Magentic-UI	Web动作与人机协作	动态站点、注入、凭证和反自动化
编排UI	vibe-kanban/HumanLayer	监督多个外部Agent	协调不替代验证和合并

原OpenCode仓库状态

GitHub快照显示 opencode-ai/opencode 已归档并且最近push停在2025-09。鉴于同名项目可能迁移或重建，本系列不把旧仓库列入36个主比较对象。使用者应核对当前官方仓库、组织和发布渠道，避免把历史仓库活跃度用于现状判断。

5. 记忆、协议与训练基础设施

仓库	层	设计	适合	
a2aproject/A2A	protocol	Protocol and SDK ecosystem for agent discovery, messaging, tasks and artifacts across vendors.	Cross-organization agent interoperability.	Protocol cor guarantee s trust.
microsoft/agent-lightning	training	Decouples agent execution and RL training through observability and trajectory transitions.	Fine-tuning existing agents without rewriting their orchestration.	Reward desi credit assigr
microsoft/Orchard	training/runtime	Kubernetes environment service and recipes for SWE, GUI and open agent harness training.	Large-scale rollout, distillation, RL and evaluation infrastructure.	Young proje operational
mem0ai/mem0	memory layer	Cross-application memory extraction, storage and retrieval service.	Personalization and persistent user facts across sessions.	Memory qua and provena per use case
getzep/graphiti	temporal graph memory	Real-time temporal knowledge graph construction and retrieval for agents.	Entity-centric histories with changing facts.	Graph extra authoritativ
letta-ai/letta	stateful agent platform	Long-lived agents with editable memory, tools and state management.	Research and products centered on persistent agent identity/state.	Long-term s and lifecycle
Alibaba-NLP/DeepResearch	research agent	Open deep-research agent and training/evaluation assets for information seeking.	Reproducible web research agents and open models.	Live search, access domi
modelcontextprotocol/servers	tool ecosystem	Reference and community MCP servers exposing external capabilities.	Learning tool contracts and bootstrapping integrations.	Community inputs and r permission r
modelcontextprotocol/modelcontextprotocol	protocol/spec	Specification and documentation for client-server context and tool exchange.	Protocol implementers and platform integration teams.	MCP standa not trust, se

记忆层选型

- 主要需求是跨会话用户事实和偏好: 从mem0类通用层起步，但补充来源和删除策略。
- 主要需求是实体关系、时间变化和冲突事实: 考察Graphiti类时序知识图，但验证抽取质量。
- 主要需求是长寿命Agent状态与可编辑记忆: 考察Letta，但先定义身份、隐私和生命周期。
- 编码/GUI任务的程序经验: 不应直接复用对话记忆指标，优先文件、事件、技能和可执行验证。

训练基础设施选型

Agent Lightning的核心是让现有Agent轨迹进入RL训练而不重写harness；Orchard的核心是把可扩展环境作为独立服务，让多个harness在隔离组件中采样、训练和评测。前者偏训练接口与信用分配，后者偏环境规模和跨harness运行。两者都说明训练不再只看模型样本，而要捕获真实Agent系统。

6. 选型决策树与组合方案

需求	候选	前置条件
只需嵌入模型+工具	OpenAI Agents SDK / PydanticAI / smolagents	选择最小抽象并自建状态/权限
需要耐久状态和人工中断	LangGraph / ADK	先做状态schema和恢复测试
需要电脑/文件/长时通用能力	Claude Agent SDK / DeepAgents	验证沙箱、压缩和模型耦合
研究多Agent通信	AutoGen / CAMEL	必须有单Agent等预算基线
业务角色流程	CrewAI / MetaGPT	让角色拥有真实不同的信息和责任
构建编码产品	OpenHands / Codex SDK类能力 / Cline	优先隔离、会话、验证和许可
Web自动化	browser-use / Magentic-UI	处理注入、登录态、动态页面和回滚
跨应用记忆	mem0 / Graphiti / Letta	按事实/关系/长寿命Agent需求选择
跨Agent互操作	A2A + MCP + Skills	分别处理Agent、工具和程序知识; 不混用信任
Agent RL	Agent Lightning + Orchard类环境	先解决oracle、rollout成本和数据版本

推荐组合A: 企业状态型Agent

PydanticAI或OpenAI Agents SDK作为轻量Agent层；LangGraph/ADK承担持久状态和工作流；MCP连接受审工具；独立权限代理签发短期凭证；事件日志和OpenTelemetry类追踪；业务规则/测试作为verifier。不要把所有能力交给一个“全能Agent”对象。

推荐组合B: 多用户编码Agent平台

OpenHands类会话和沙箱平台，或自建Agent SDK + 容器/微VM；Git分支作为隔离状态；任务队列和checkpoint；测试/静态分析/视觉检查；HumanLayer或任务看板做监督；模型适配层可切换；秘密通过代理注入且不可直接读取。

推荐组合C: 开放研究Agent

DeepResearch/smolagents类可读harness；搜索、浏览、代码和引用工具；证据表作为共享状态；独立检索和写作Agent；BrowseComp-Plus/DeepSearchQA类任务评测；网页内容默认不可信；所有结论保留局部引用。

7. 36个仓库逐项研究卡

R01 | openai/openai-agents-python | SDK/runtime
快照: Python; MIT; 28,773 stars; created 2025-03; active
设计: Small agent loop with agents, tools, handoffs, sessions, guardrails and tracing.
适合: OpenAI-centric service agents and multi-agent workflows.
注意: Provider coupling and rapidly moving platform APIs require version pinning.
GitHub

R02 | openai/openai-agents-js | SDK/runtime
快照: TypeScript; MIT; 3,659 stars; created 2025-05; active
设计: TypeScript SDK with multi-agent and realtime voice support.
适合: Web, Node and voice applications.
注意: Smaller ecosystem than the Python package and fast API churn.
GitHub

R03 | openai/codex | coding agent
快照: Rust; Apache-2.0; 106,864 stars; created 2025-04; active
设计: Terminal coding harness with sandboxed command execution, repository context and iterative edits.
适合: Local and remote software engineering workflows.
注意: Model-specific behavior and large operational surface; measure full harness, not model alone.
GitHub

R04 | anthropics/claude-code | coding agent
快照: Python; proprietary/see repo; 142,023 stars; created 2025-02; active
设计: Terminal agent with file, shell, git, subagent, hook, skill and compaction primitives.
适合: Long-running coding and general computer work.
注意: Client repository visibility does not expose the full hosted model or control plane.
GitHub

R05 | anthropics/claude-agent-sdk-python | SDK/harness
快照: Python; MIT; 7,935 stars; created 2025-06; active
设计: Generalizes Claude Code's harness primitives to application agents.
适合: Agents that need a computer, subagents, hooks and permission controls.
注意: Closely tied to Claude behaviors and platform evolution.
GitHub

R06 | google/adk-python | SDK/runtime
快照: Python; Apache-2.0; 21,194 stars; created 2025-04; active
设计: Code-first agents, tools, sessions, memory, artifacts, evaluation and deployment.
适合: Google Cloud/Gemini teams needing multi-agent and managed deployment.
注意: Broad abstraction surface can obscure runtime cost and state ownership.
GitHub

R07 | google/adk-js | SDK/runtime
快照: TypeScript; Apache-2.0; 1,354 stars; created 2025-08; active
设计: TypeScript implementation of ADK's agent and deployment abstractions.
适合: Node teams aligning with ADK architecture.
注意: Newer and less mature than the Python implementation.
GitHub

R08 | a2aproject/A2A | protocol
快照: Shell/spec; Apache-2.0; 25,423 stars; created 2025-03; active
设计: Protocol and SDK ecosystem for agent discovery, messaging, tasks and artifacts across vendors.
适合: Cross-organization agent interoperability.
注意: Protocol compatibility does not guarantee semantic capability or trust.
GitHub

R09 | microsoft/autogen | multi-agent framework
快照: Python; CC-BY-4.0 metadata; 60,525 stars; created 2023-08; active
设计: AgentChat abstractions over an event-driven core and extensions.
适合: Research and conversational multi-agent orchestration.
注意: Large version transitions and licensing/component details need review.
GitHub

R10 | microsoft/agent-lightning | training
快照: Python; MIT; 17,520 stars; created 2025-06; active
设计: Decouples agent execution and RL training through observability and trajectory transitions.
适合: Fine-tuning existing agents without rewriting their orchestration.
注意: Reward design, rollout cost and credit assignment remain difficult.
GitHub

R11 | microsoft/magentic-ui | computer-use
快照: Python; MIT; 10,069 stars; created 2025-05; active
设计: Human-centered browser and file agent built on AutoGen/Magentic patterns.
适合: Research on oversight and interactive web automation.
注意: Live web fragility and browser security remain central.
GitHub

R12 | microsoft/Orchard | training/runtime
快照: mixed; MIT; 478 stars; created 2026-04; active
设计: Kubernetes environment service and recipes for SWE, GUI and open agent harness training.
适合: Large-scale rollout, distillation, RL and evaluation infrastructure.
注意: Young project with meaningful operational complexity.
GitHub

R13 | langchain-ai/langgraph | durable runtime
快照: Python; MIT; 40,040 stars; created 2023-08; active
设计: Graph/state runtime with durable execution, persistence, streaming and human-in-the-loop.
适合: Stateful production workflows requiring recovery and explicit control flow.
注意: Graph complexity can grow into application-specific orchestration debt.
GitHub

R14 | langchain-ai/deepagents | agent harness
快照: Python; MIT; 27,949 stars; created 2025-07; active
设计: Batteries-included harness adding planning, file state, subagents and memory on LangGraph.
适合: Long-horizon research and coding prototypes.
注意: Convenience layers inherit dependencies and defaults from the LangChain stack.
GitHub

R15 | crewAIInc/crewAI | multi-agent framework
快照: Python; MIT; 57,338 stars; created 2023-10; active
设计: Role/task/crew and flow abstractions for collaborative automation.
适合: Business workflows where roles and sequence are easy to express.
注意: Role-playing can become cosmetic unless information and responsibility boundaries are real.
GitHub

R16 | pydantic/pydantic-ai | typed SDK
快照: Python; MIT; 19,399 stars; created 2024-06; active
设计: Typed model and tool interfaces integrated with validation, evals and observability.
适合: Python teams prioritizing type safety and model portability.
注意: A typed boundary catches schema errors, not semantic or planning failures.
GitHub

R17 | huggingface/smolagents | minimal SDK
快照: Python; Apache-2.0; 28,890 stars; created 2024-12; active
设计: Minimal agents that often express plans and tool composition as code.
适合: Readable experiments and code-agent research.
注意: Small core does not provide a production control plane or isolation by itself.
GitHub

R18 | browser-use/browser-use | web agent
快照: Python; MIT; 109,782 stars; created 2024-10; active
设计: Browser abstraction plus agent loop over Playwright-like interactions.
适合: Web automation and browser-agent prototyping.
注意: Site change, anti-bot behavior, credentials and prompt injection create operational risk.
GitHub

R19 | OpenHands/OpenHands | coding platform
快照: TypeScript; MIT; 84,512 stars; created 2024-03; active
设计: Full coding-agent platform with isolated runtime, sessions, UI and extensible agents.
适合: Multi-user hosted software-development agents.
注意: Deployment complexity is much larger than a local CLI harness.
GitHub

R20 | SWE-agent/SWE-agent | coding/eval
快照: Python; MIT; 20,079 stars; created 2024-04; active
设计: Research harness for GitHub issue resolution and other terminal tasks.
适合: Transparent coding-agent research and benchmark baselines.
注意: Benchmark performance is sensitive to environment and task contamination.
GitHub

R21 | mem0ai/mem0 | memory layer
快照: Python; Apache-2.0; 63,619 stars; created 2023-06; active
设计: Cross-application memory extraction, storage and retrieval service.
适合: Personalization and persistent user facts across sessions.
注意: Memory quality, deletion semantics and provenance must be evaluated per use case.
GitHub

R22 | getzep/graphiti | temporal graph memory
快照: Python; Apache-2.0; 30,104 stars; created 2024-08; active
设计: Real-time temporal knowledge graph construction and retrieval for agents.
适合: Entity-centric histories with changing facts.
注意: Graph extraction errors can appear authoritative and propagate widely.
GitHub

R23 | letta-ai/letta | stateful agent platform
快照: mixed; Apache-2.0; 24,307 stars; created 2023-10; active
设计: Long-lived agents with editable memory, tools and state management.
适合: Research and products centered on persistent agent identity/state.
注意: Long-term state creates privacy, drift and lifecycle obligations.
GitHub

R24 | agno-agi/agno | agent platform
快照: Python; Apache-2.0; 41,790 stars; created 2022-05; active
设计: Broad toolkit for building, running and managing agents and teams.
适合: Teams wanting many integrations under one Python platform.
注意: Breadth increases dependency and abstraction risk; choose a minimal subset.
GitHub

R25 | camel-ai/camel | multi-agent research
快照: Python; Apache-2.0; 17,610 stars; created 2023-03; active
设计: Communicative-agent and society abstractions plus research environments.
适合: Multi-agent research, simulations and data generation.
注意: Simulation success does not imply robust operational collaboration.
GitHub

R26 | QwenLM/Qwen-Agent | SDK/app
快照: Python; Apache-2.0; 16,992 stars; created 2023-09; active
设计: Function calling, MCP, code interpreter, RAG and browser extensions for Qwen models.
适合: Qwen-centric local or enterprise agent applications.
注意: Recent push cadence was lower at the snapshot; verify current compatibility.
GitHub

R27 | Alibaba-NLP/DeepResearch | research agent
快照: Python; Apache-2.0; 19,851 stars; created 2025-01; active
设计: Open deep-research agent and training/evaluation assets for information seeking.
适合: Reproducible web research agents and open models.
注意: Live search, citation quality and site access dominate practical reliability.
GitHub

R28 | modelcontextprotocol/servers | tool ecosystem
快照: TypeScript; see repo; 89,696 stars; created 2024-11; active
设计: Reference and community MCP servers exposing external capabilities.
适合: Learning tool contracts and bootstrapping integrations.
注意: Community tools are supply-chain inputs and require provenance and permission review.
GitHub

R29 | modelcontextprotocol/modelcontextprotocol | protocol/spec
快照: TypeScript; see repo; 8,998 stars; created 2024-09; active
设计: Specification and documentation for client-server context and tool exchange.
适合: Protocol implementers and platform integration teams.
注意: MCP standardizes transport/shape, not trust, semantics or policy.
GitHub

R30 | FoundationAgents/MetaGPT | multi-agent framework
快照: Python; MIT; 69,901 stars; created 2023-06; active
设计: Software-company role workflow and natural-language programming patterns.
适合: Educational multi-agent software-process experiments.
注意: Role simulation and benchmark claims need controlled comparison with simpler agents.
GitHub

R31 | Aider-AI/aider | coding agent
快照: Python; Apache-2.0; 48,333 stars; created 2023-05; active
设计: Git-aware terminal pair programmer with repository maps and edit formats.
适合: Interactive coding with explicit human steering.
注意: Less suited to unattended multi-user control-plane workloads.
GitHub

R32 | cline/cline | coding agent
快照: TypeScript; Apache-2.0; 66,499 stars; created 2024-07; active
设计: IDE, CLI and SDK agent with file, terminal, browser and approval flows.
适合: Developer-in-the-loop work inside editors.
注意: Extension permissions and provider configuration expand the attack surface.
GitHub

R33 | continuedev/continue | coding agent
快照: TypeScript; Apache-2.0; 35,546 stars; created 2023-05; active
设计: Open coding agent and editor integrations with model/provider flexibility.
适合: Teams needing self-hosted or model-flexible coding assistance.
注意: Configuration flexibility transfers integration and evaluation work to the operator.
GitHub

R34 | google-gemini/gemini-cli | coding/general agent
快照: TypeScript; Apache-2.0; 106,586 stars; created 2025-04; active
设计: Terminal agent with Gemini models, tools, MCP and extensions.
适合: Google-centric development and general terminal workflows.
注意: Provider coupling and broad terminal permissions need explicit containment.
GitHub

R35 | BloopAI/vibe-kanban | orchestrator/UI
快照: Rust; Apache-2.0; 27,856 stars; created 2025-06; active
设计: Task-board orchestration over multiple external coding agents.
适合: Human supervision of parallel coding work.
注意: It coordinates harnesses but does not eliminate merge, verification or context problems.
GitHub

R36 | humanlayer/humanlayer | coding workflow
快照: TypeScript; see repo; 11,296 stars; created 2024-08; active
设计: Human-in-the-loop workflow and context engineering for complex codebases.
适合: High-context changes needing planning and review checkpoints.
注意: Process quality depends on disciplined task decomposition and review capacity.
GitHub

本卷结论
选Agent仓库时，先确定你需要的是循环SDK、耐久运行时、完整harness、领域应用、协议、记忆还是训练环境。再以真实任务、恢复、安全和单位成功成本验证。仓库stars与功能数量只能帮助发现候选，不能替代工程尽调。